

COVID-19's Impact on Motor Vehicle Collisions in NYC

STAT 605 Final Presentation

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Introduction

Project Overview

Research Question: How did COVID-19 affect motor vehicle collision patterns in New York City?

- **Dataset:** 2.22M police-reported crashes (2012-2025)
- **Time Periods:**
 - Pre-COVID (before March 2020)
 - During COVID (March 2020 - Dec 2021)
 - Post-COVID (2022-present)
- **Key Questions:**
 1. Did collision rates return to pre-pandemic levels?
 2. Did crashes become more deadly?
 3. What changed in driver behavior?

Data Sources

Datasets:

- **Primary:** NYC Motor Vehicle Collisions (2.22M records)
- **Secondary:** Vehicle-Level Details (4.45M records)
- **Tertiary:** NYC Speed Limits (141K segments)
- **Quaternary:** Twitter Traffic Reports (51K tweets)

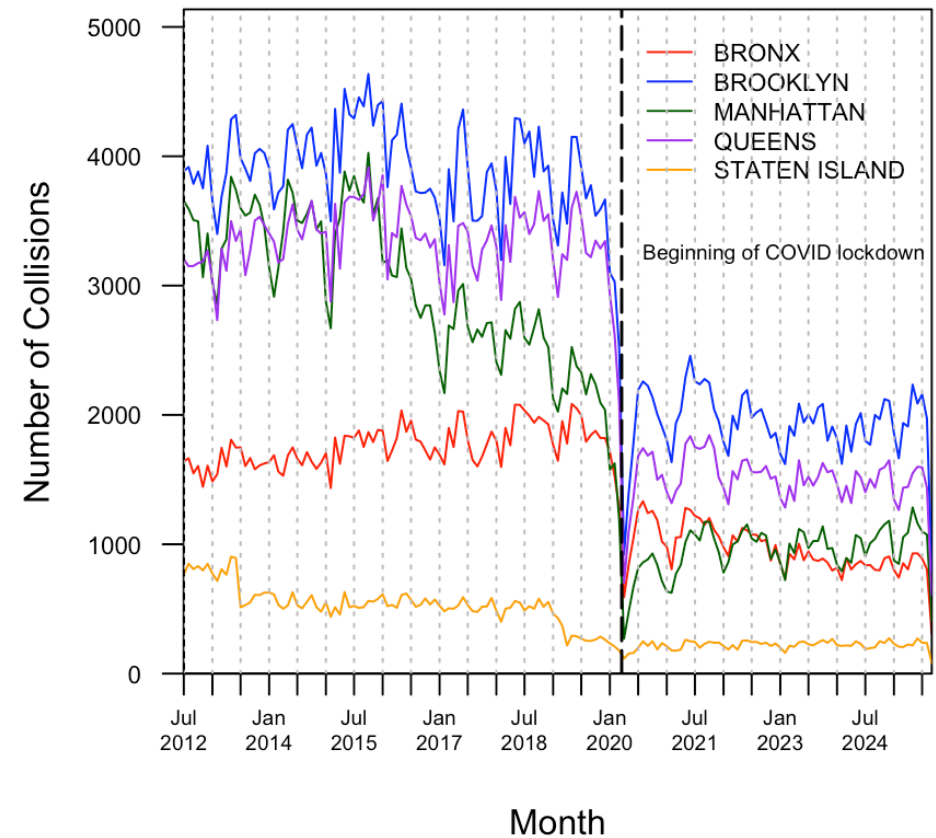
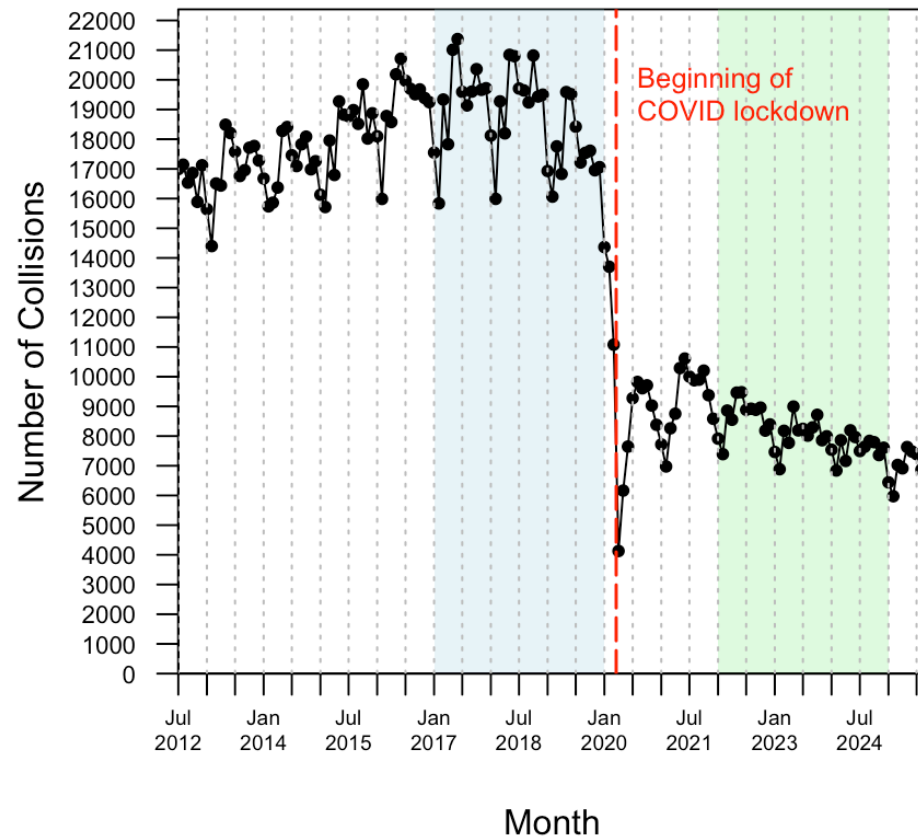
Key Variables:

- Crash date/time & location
- Casualties by road user type
- Contributing factors (up to 5)
- Vehicle characteristics
- Borough information

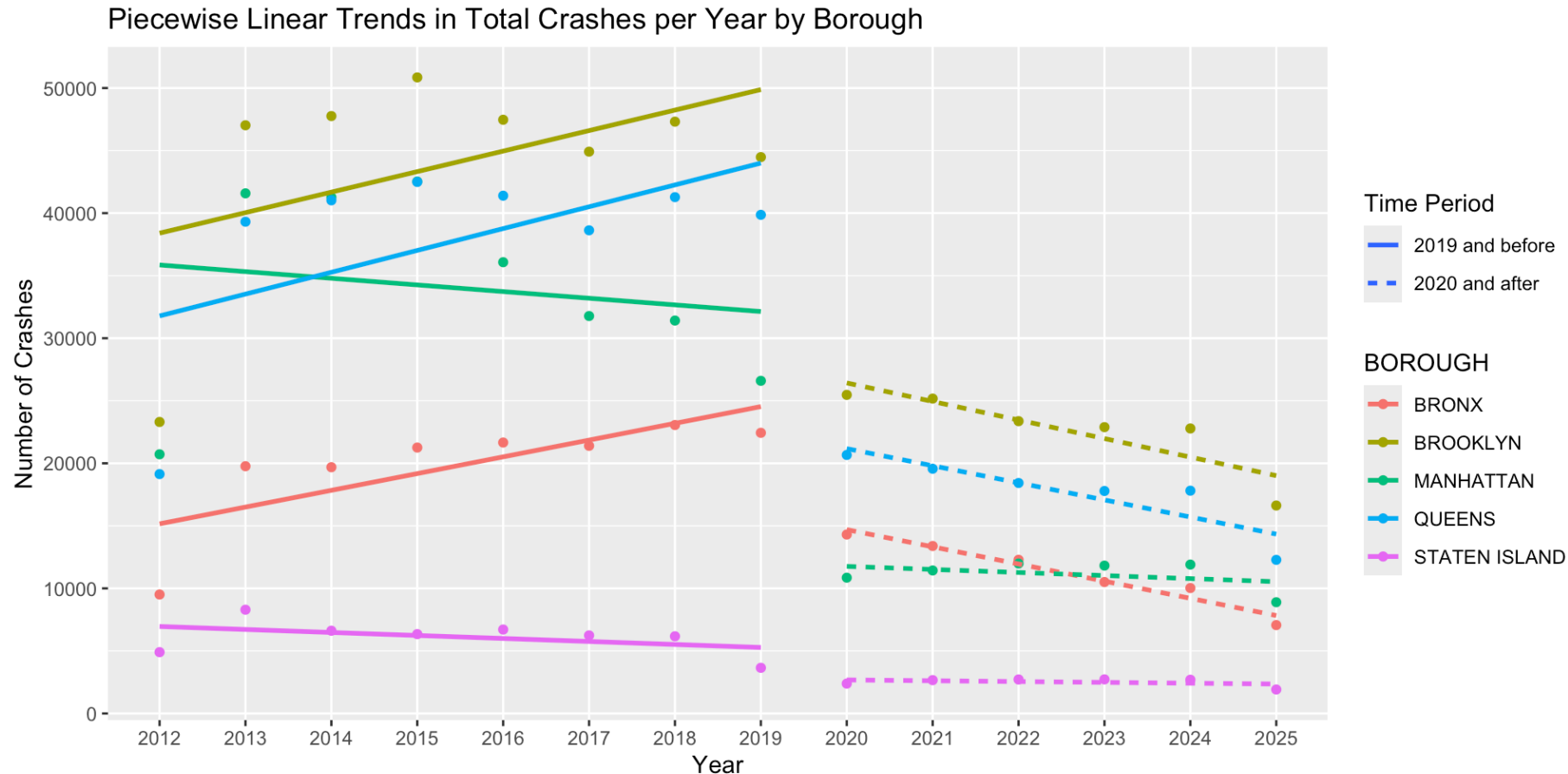
Did Collision Rates Change?

Monthly Collision Trends

Number of Monthly Vehicle Collisions in NYC:



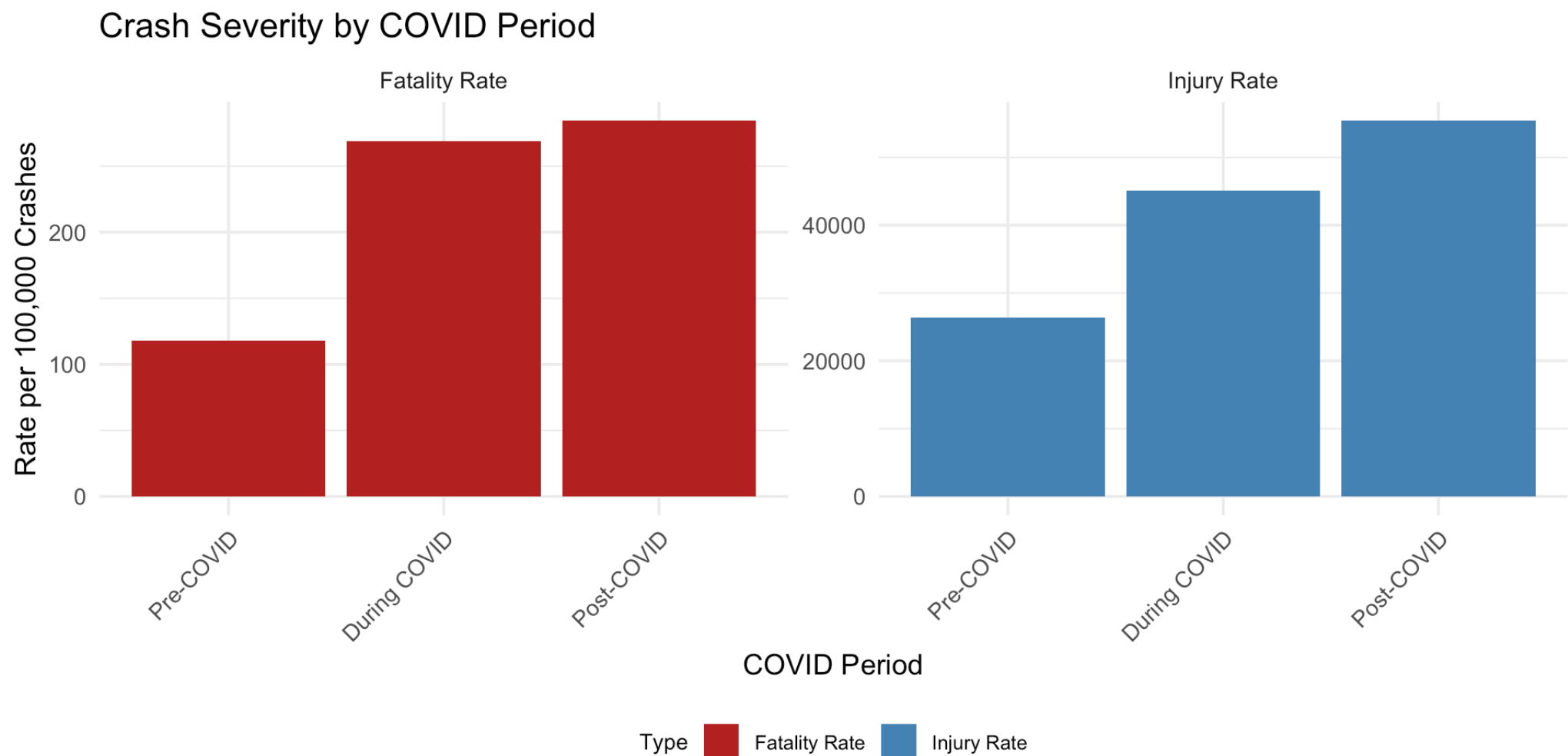
Borough-Level Patterns



- **Brooklyn & Queens & Bronx:** Increasing pre-COVID, Decreasing post-COVID
- **Manhattan:** Decreasing pre-COVID, more static post-COVID

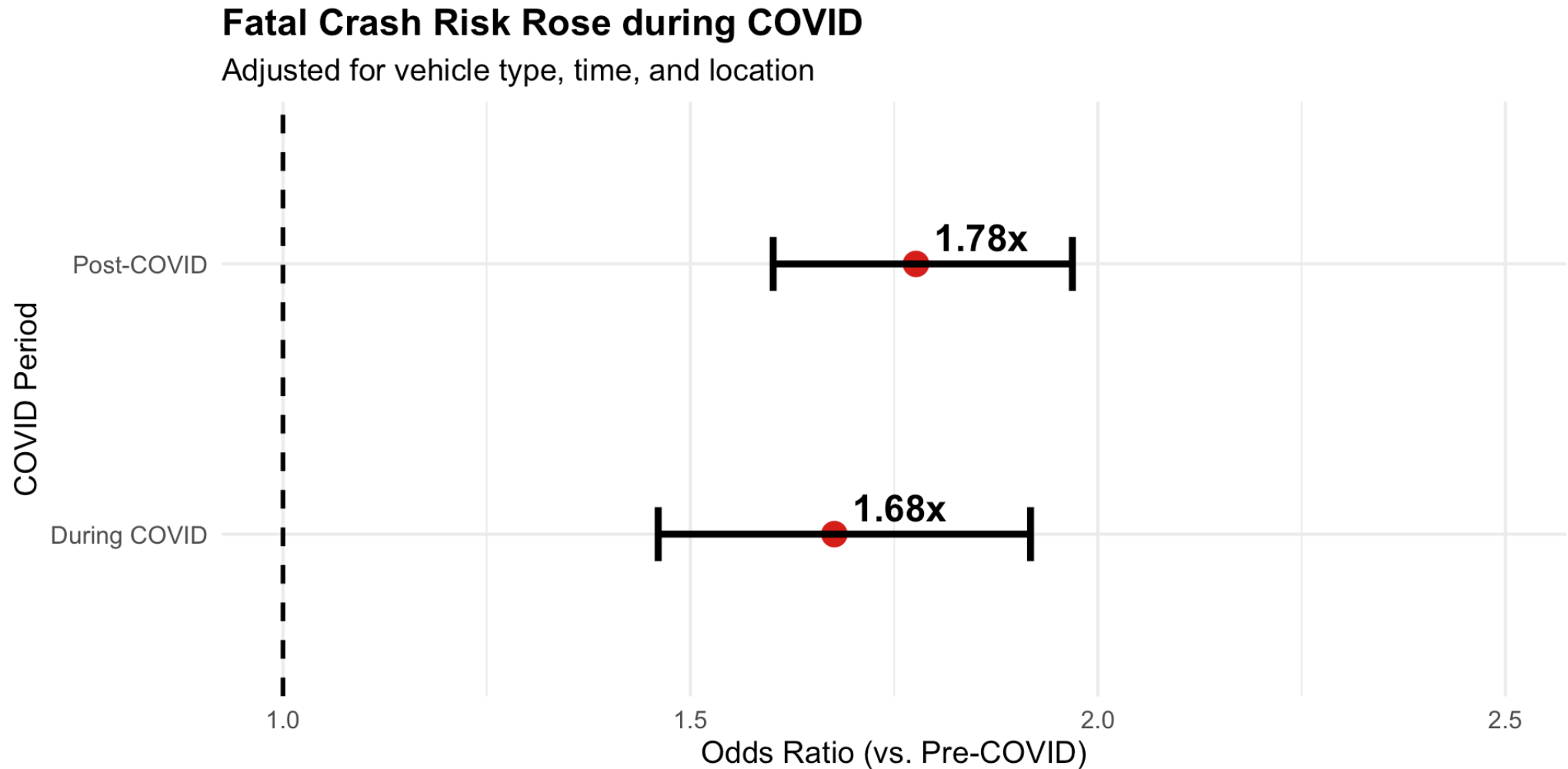
Did Crashes Become More Severe?

Severity



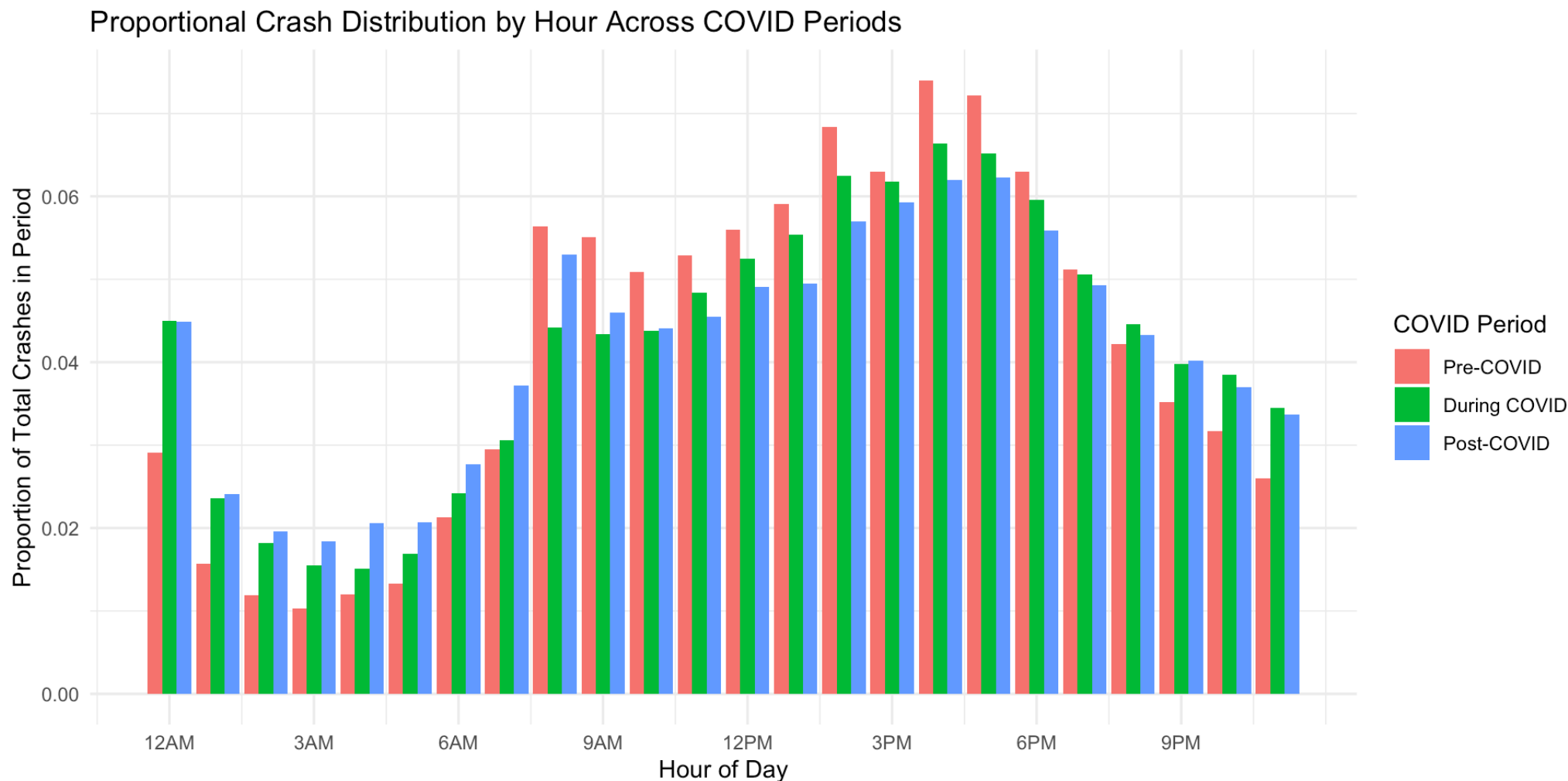
Key Finding: Crash severity increased during and after COVID-19

COVID's Impact on Severity



Key Finding: Even controlling for other factors, crashes became 67-78% more likely to be fatal

Crash Timing Patterns

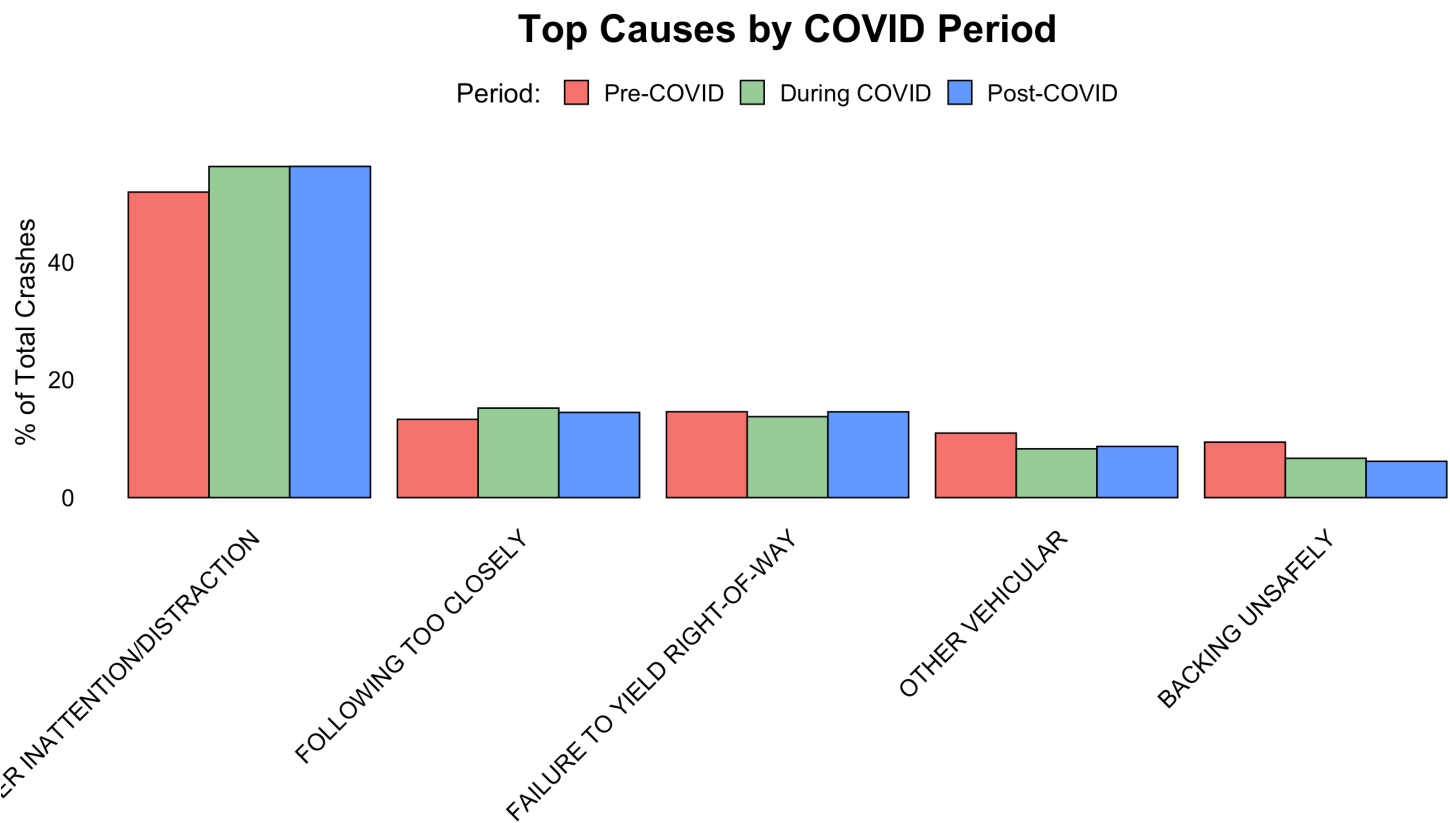


Rush-hour peaks dipped slightly during COVID but largely returned afterward

What Caused These Changes?

Contributing Factors

How Factors Changed:



Key Finding: Driver inattention and distraction are the leading cause and became more common during COVID

Text Mining Analysis

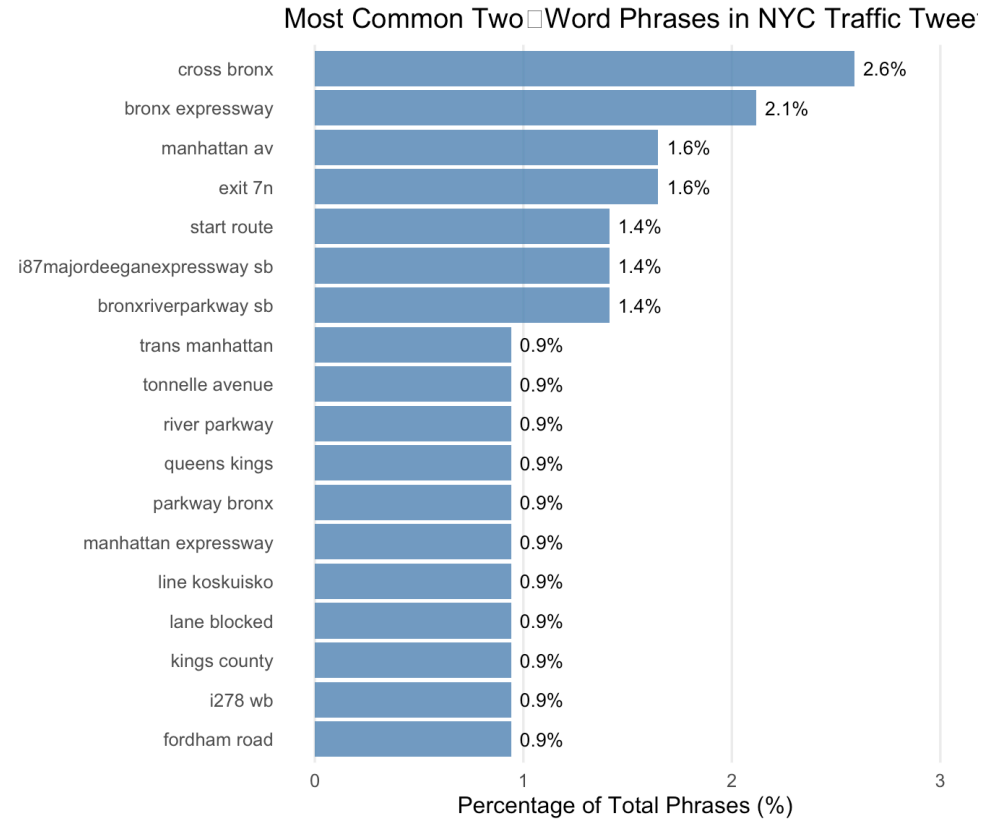
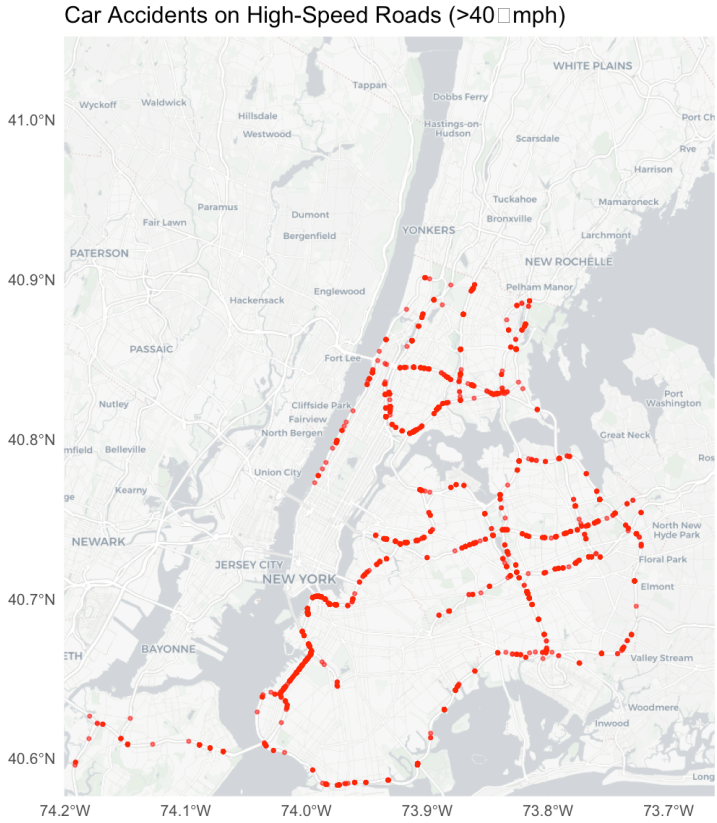
Word Cloud

Data Processing:

- Started with 51K traffic-related tweets
- Filtered to NYC-specific mentions
- Isolated vehicle crashes (excluded transit)
- Final dataset: **700+ vehicle crash tweets from 2018**

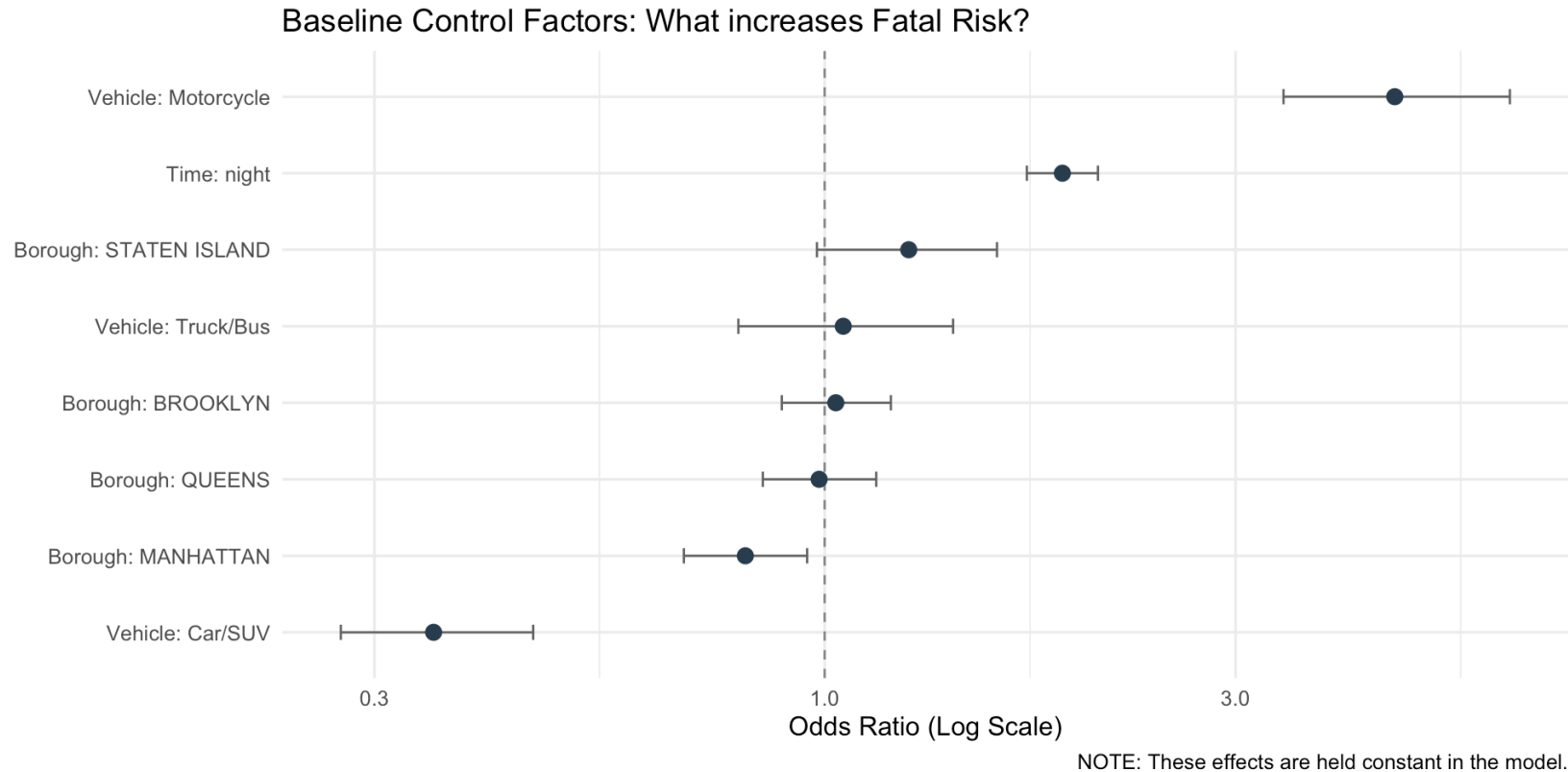


Top Phrases



Statistical Modeling

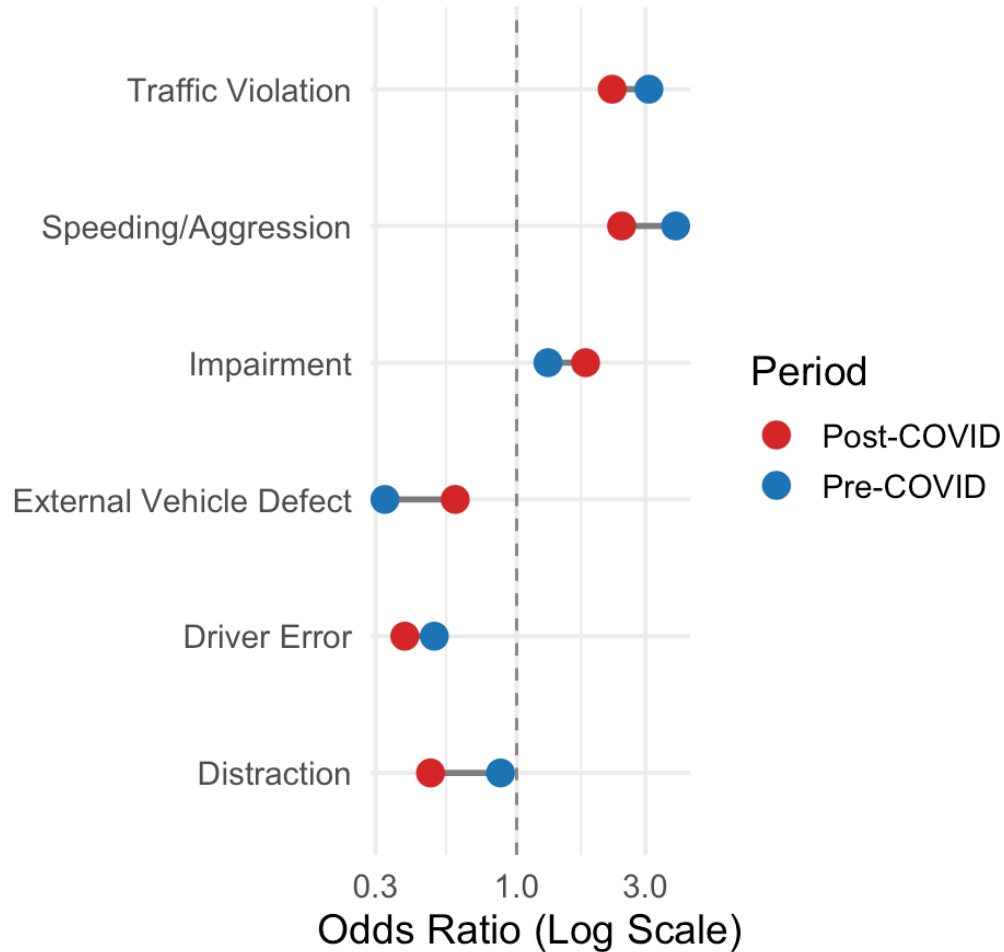
Baseline Risk Factors



- **Motorcycles:** 3x+ more likely to be fatal
- **Nighttime:** ~2x higher odds than daytime
- **Manhattan:** Lower risk due to congestion

How COVID Changed Risk

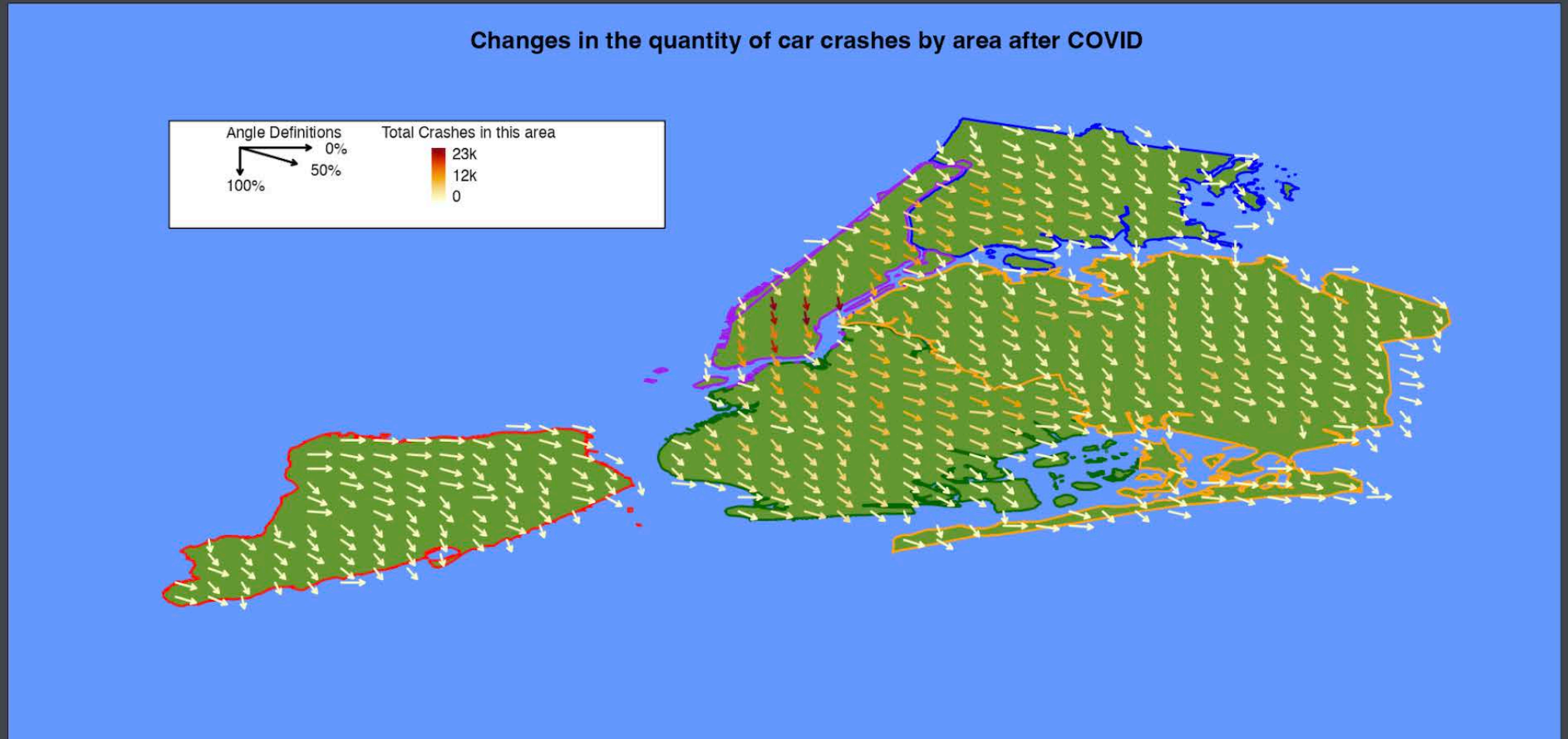
How Did Fatal Crash Risks Shift?



Did the Risk Change Significantly?



Killer Plot



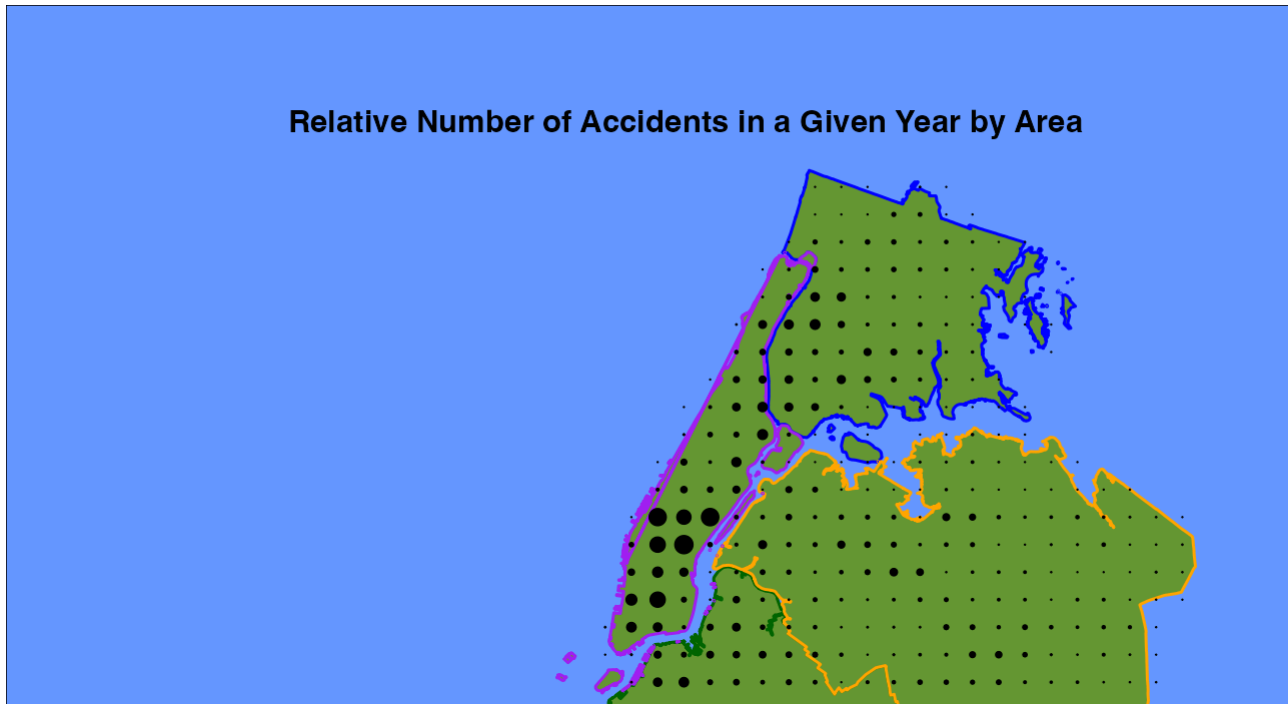
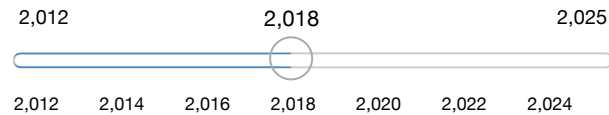
Interactive Killer Plot

Adjustable Map

Choose view:

- ☒ Accident Distribution
☐ Accident Trends

Year:



Conclusion

Key Findings

1. Collision trends

- Rates stayed lower than before COVID and never fully returned
- Borough distribution: Brooklyn and Queens dropped the most and recovered slowly; Manhattan rebounded fastest; the Bronx was intermediate

2. Severity trends

- Fatality rate per crash rose sharply during and after COVID
- Injuries per crash fell, indicating fewer but more lethal crashes

3. Driver-behavior

- Speeding/Aggression and Impairment remained high-risk with unchanged odds
- Distraction and Traffic-Violation odds fell significantly

Thank You!

